

Computer Networks & Networking

FIG_000 · guide v1.0 · 2026 · open reference · 16 chapters
by Swastik Sagar

Concepts, architectures, protocols, and diagrams – from first principles to real-world network design. Every chapter opens into a full explanation, worked examples, and a labeled diagram.

```
$ read chapter_01 --start
```

WHAT'S INSIDE

PART	CHAPTERS	FOCUS
I · Foundations	1 – 5	Network types, topologies, the OSI & TCP/IP models
II · Core Mechanics	6 – 10	Media, devices, switching, IP addressing, core protocols
III · Reliability & Routing	11 – 13	Error handling, flow/congestion control, routing
IV · Applied & Secure	14 – 16	Wireless networking, security fundamentals, glossary

How to use this guide: each chapter is self-contained – read start to finish for a full course, or jump straight to the chapter you need using the Table of Contents. Every diagram is numbered and referenced directly in the surrounding text.

Table of Contents

1. Introduction to Computer Networks	3
2. Network Types & Scale	5
3. Network Topologies	7
4. The OSI Reference Model	10
5. The TCP/IP Model	12
6. Transmission Media	13
7. Networking Devices	14
8. Switching Techniques	16
9. IP Addressing & Subnetting	17
10. Core Protocols (TCP, UDP, HTTP, DNS, DHCP...)	19
11. Error Detection & Correction	21
12. Flow Control & Congestion Control	22
13. Routing Algorithms & Protocols	23
14. Wireless Networking	24
15. Network Security Fundamentals	25
16. Glossary of Key Terms	27

This guide is designed to be read chapter by chapter, with diagrams positioned next to the concepts they illustrate.

Introduction to Computer Networks

1.1 WHAT IS A COMPUTER NETWORK?

A **computer network** is a collection of two or more interconnected computing devices – computers, servers, smartphones, printers, sensors – that can exchange data and share resources with one another using a common set of rules called **protocols**. The connection between devices can be established using physical cables (wired) or radio signals (wireless).

Networks exist so that isolated machines don't have to work alone. Instead of copying a file onto a USB drive and walking it across the office, a network lets that file travel over a wire or through the air in milliseconds. This simple idea – letting machines talk to each other – is the foundation of everything from email to streaming video to the entire Internet.

Key idea: A network is defined by three things – the **nodes** (devices), the **links** (the medium connecting them), and the **protocols** (the rules governing communication). Remove any one of these and you no longer have a functioning network.

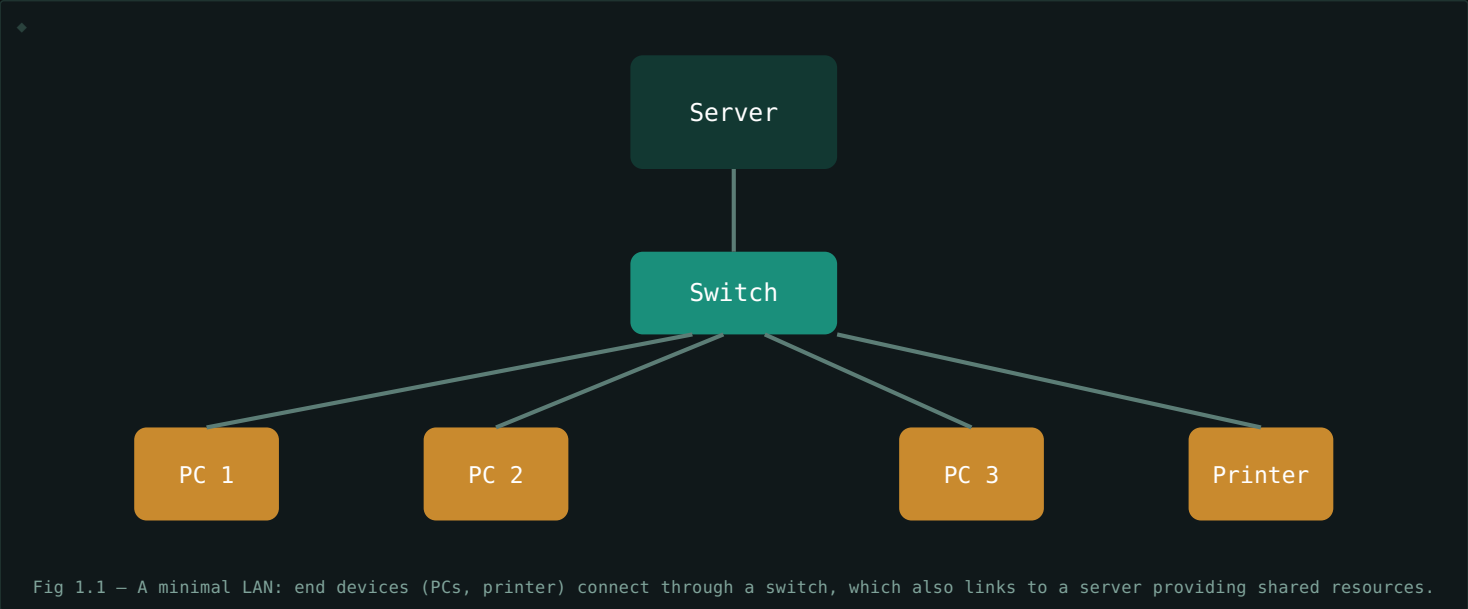
1.2 WHY DO WE NEED NETWORKS?

- **Resource Sharing** – multiple users can share a single printer, scanner, internet connection, or storage server.
- **Communication** – email, chat, video calls, and file transfer all depend on networks.
- **Data Centralization** – files and databases can live on a central server, so data stays consistent and backed up.
- **Scalability & Cost Efficiency** – expensive hardware (servers, high-speed printers) can be shared instead of duplicated per user.
- **Reliability** – data can be replicated across multiple machines/locations so a single failure doesn't take everything down.

1.3 BASIC COMPONENTS OF A NETWORK

COMPONENT	ROLE
Node / Host	Any device that sends or receives data – PC, server, phone, IoT sensor, printer.
Link / Medium	The physical or wireless path data travels on – copper cable, fiber, radio waves.
Network Interface Card (NIC)	Hardware that connects a device to the network medium and gives it a unique MAC address.
Protocol	Agreed-upon rules that define how data is formatted, addressed, transmitted, and received (e.g. TCP/IP, HTTP).
Intermediary Devices	Hubs, switches, routers, and access points that forward, filter, or direct traffic between nodes.

1.4 A SIMPLE NETWORK PICTURE



1.5 ADVANTAGES & DISADVANTAGES OF NETWORKING

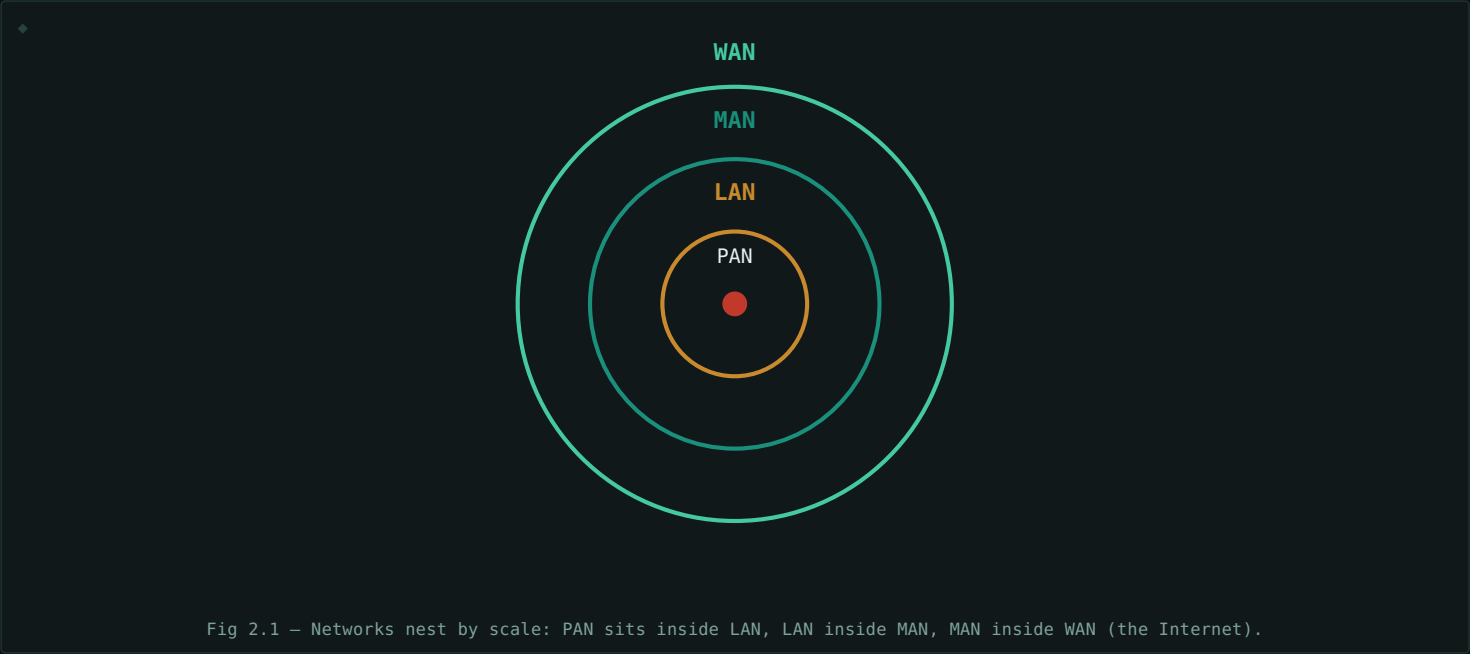
ADVANTAGES	DISADVANTAGES
Easy resource and file sharing	Security risks – more entry points for attackers
Centralized data management and backups	Setup and maintenance cost
Faster, cheaper communication	Dependency – network failure disrupts all connected work
Scalable as an organization grows	Requires skilled administration

Network Types & Scale

Networks are usually classified by the geographical area they cover. As the distance grows, the technology, ownership model, and speed characteristics change.

2.1 CLASSIFICATION BY GEOGRAPHIC SPAN

TYPE	FULL FORM	TYPICAL RANGE	EXAMPLE
PAN	Personal Area Network	Up to ~10 m	Bluetooth earphones connected to a phone
LAN	Local Area Network	A room / building / campus	Office or home Wi-Fi network
MAN	Metropolitan Area Network	A city	Cable TV network across a city
WAN	Wide Area Network	Country / continent / global	The Internet itself
CAN	Campus Area Network	A university/corporate campus	Multiple buildings on one campus interconnected



Network Types & Scale

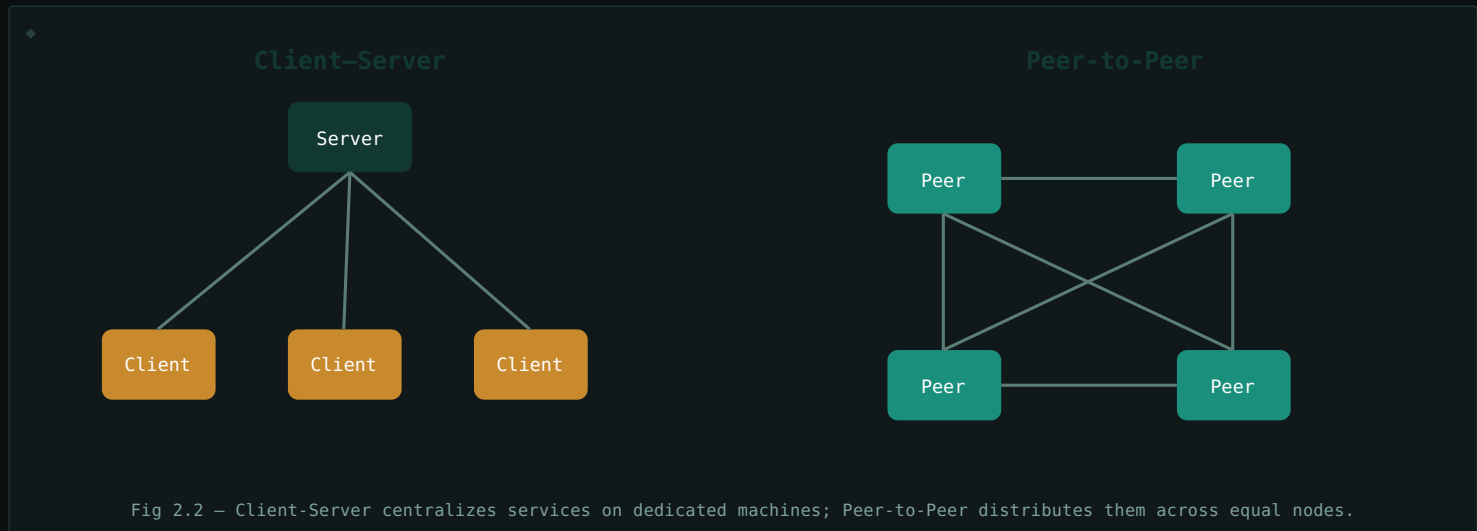
2.2 CLASSIFICATION BY ARCHITECTURE

Client-Server Network

Dedicated powerful machines called **servers** provide services (files, email, web pages, databases) to many **client** machines that request them. Centralized control makes security and backup easier, but the server is a potential single point of failure and needs dedicated hardware.

Peer-to-Peer (P2P) Network

Every device is both a client and a server – each node can share and request resources directly from others, with no central authority. Cheaper and simple to set up, but harder to secure and manage at scale (used in torrenting, some blockchain networks, and small home networks).



Network Topologies

A **topology** is the arrangement – physical or logical – of nodes and links in a network. The choice of topology affects cost, reliability, scalability, and ease of troubleshooting.

3.1 BUS TOPOLOGY

All devices are connected to a single central cable, called the **backbone** or **bus**. Data sent by any device travels along the bus in both directions until it reaches its destination.



Fig 3.1 – Bus topology: every node taps into one shared backbone cable (terminators at each end shown in red).

Pros: cheap, simple to install for small networks. **Cons:** a break anywhere on the backbone brings the whole network down; performance degrades as more nodes are added.

3.2 STAR TOPOLOGY

All devices connect individually to a central device – typically a switch or hub. This is the most common topology in modern LANs.

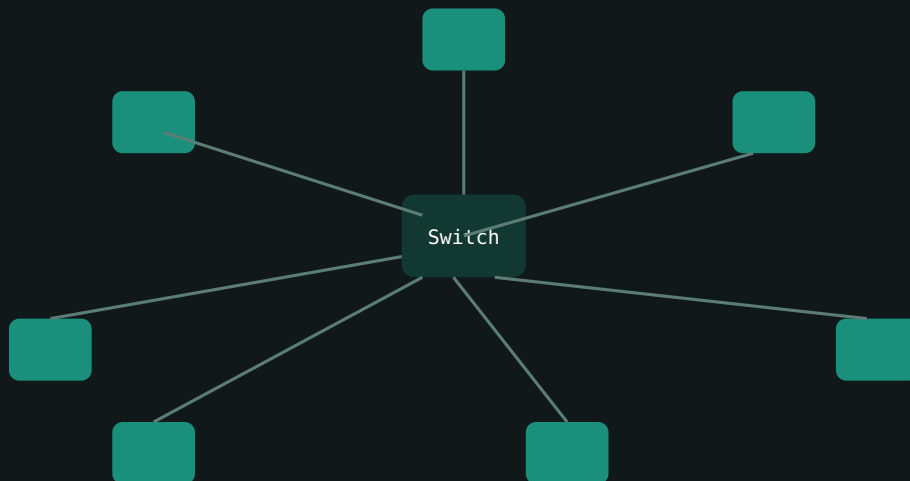


Fig 3.2 – Star topology: every node has its own dedicated link back to a central switch.

Pros: a single cable failure only affects one node; easy to add/remove devices; centralized troubleshooting. **Cons:** the central device is a single point of failure; requires more cabling than bus.

3.3 RING TOPOLOGY

Each device connects to exactly two neighbors, forming a closed loop. Data travels around the ring, typically in one direction, until it reaches its destination.

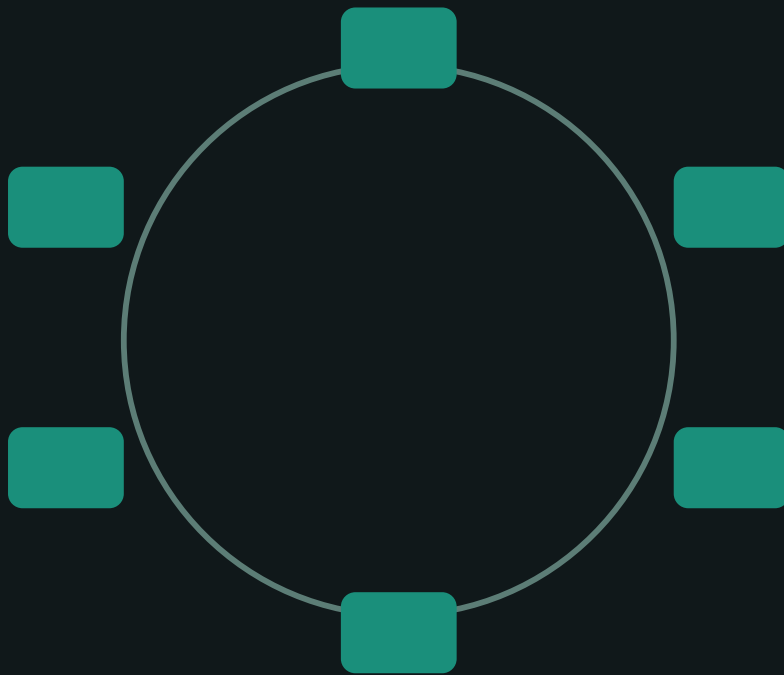


Fig 3.3 – Ring topology: data hops from node to node around a closed loop.

Pros: predictable performance, no collisions in token-based rings. **Cons:** a single broken link/node can disrupt the whole ring (unless a dual/redundant ring is used).

3.4 MESH TOPOLOGY

Every device connects directly to every other device. In a **full mesh** with n nodes, there are $\frac{n(n-1)}{2}$ links.

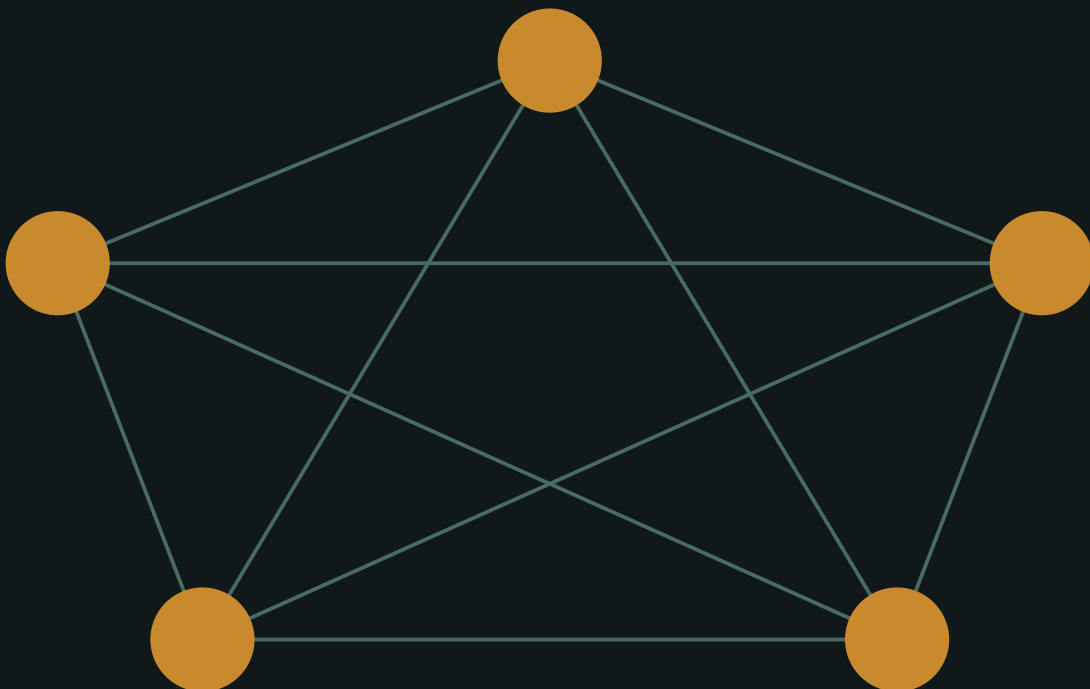
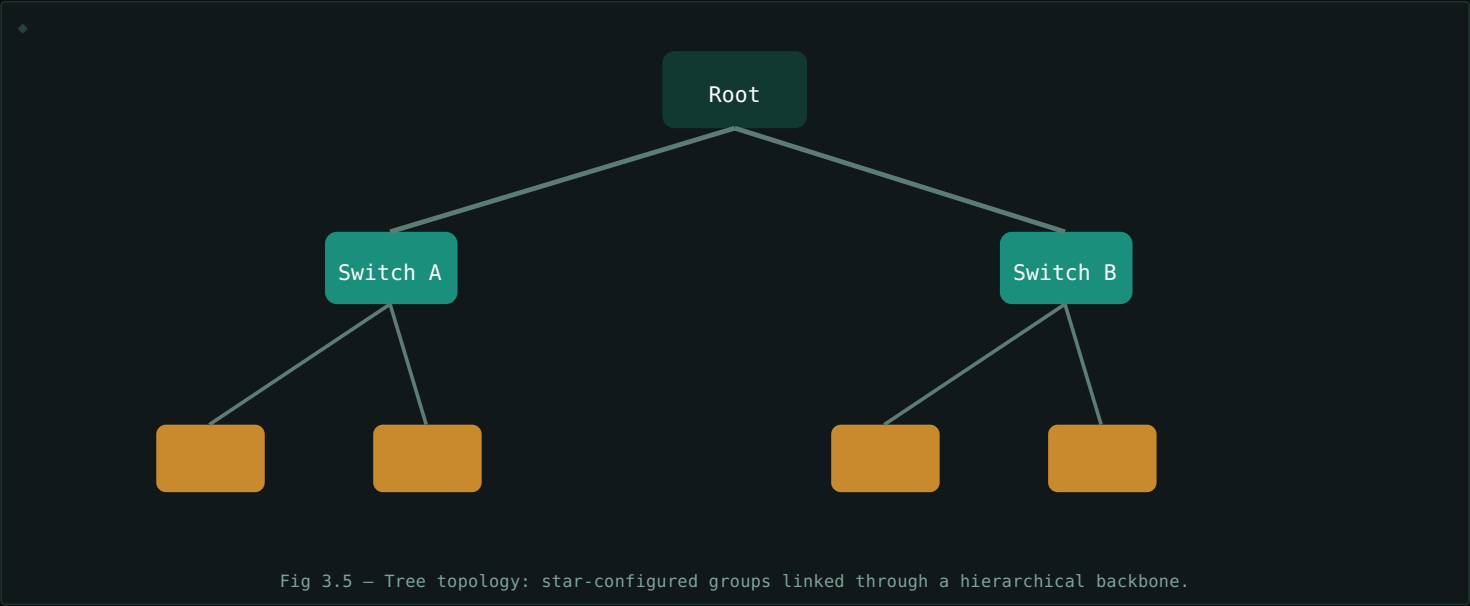


Fig 3.4 – Full mesh topology: 5 nodes require 10 dedicated links, giving maximum redundancy.

Pros: extremely fault-tolerant – no single point of failure; high privacy per link. **Cons:** expensive and complex to wire; doesn't scale well (cabling grows quadratically).

3.5 TREE (HIERARCHICAL) TOPOLOGY

A hybrid of bus and star: groups of star networks are connected to a central bus, forming a hierarchy – much like an organization chart. Common in large campus networks with a core, distribution, and access layer.



3.6 HYBRID TOPOLOGY

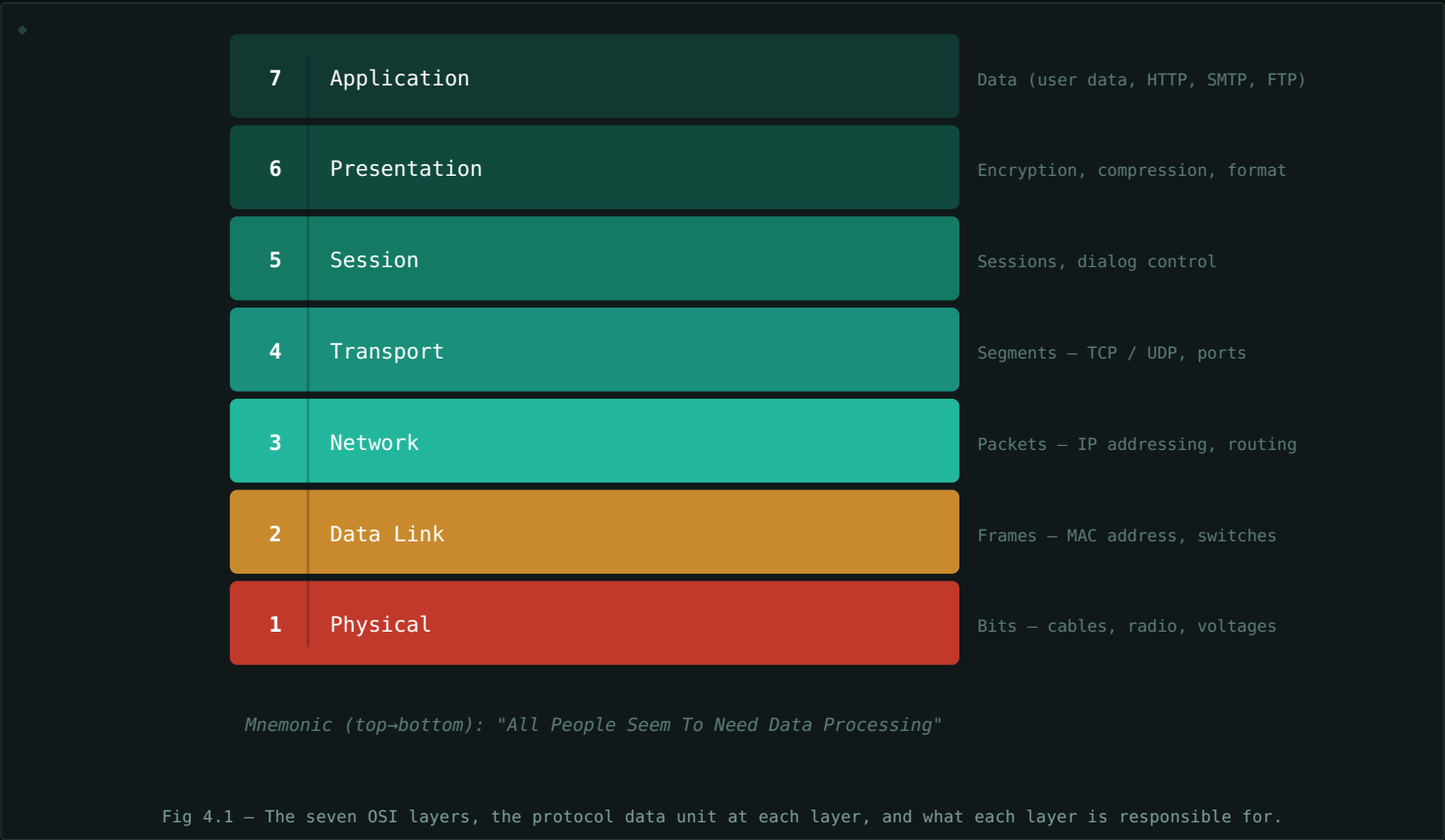
A real-world combination of two or more basic topologies – e.g., a star-of-rings, or a star network connected to a mesh backbone for redundancy. Most large enterprise and ISP networks are hybrids, chosen to balance cost, performance, and fault tolerance.

3.7 TOPOLOGY COMPARISON

TOPOLOGY	FAULT TOLERANCE	COST	SCALABILITY	TYPICAL USE
Bus	Low	Low	Poor	Small, legacy Ethernet segments
Star	Medium	Medium	Good	Modern LANs / office networks
Ring	Low–Medium	Medium	Medium	Token Ring, some MANs (FDDI)
Mesh	Very High	High	Poor (cost grows fast)	Critical backbones, military/ISP cores
Tree	Medium	Medium	Very Good	Campus & enterprise networks
Hybrid	High	Varies	Very Good	Real-world large-scale networks

The OSI Reference Model

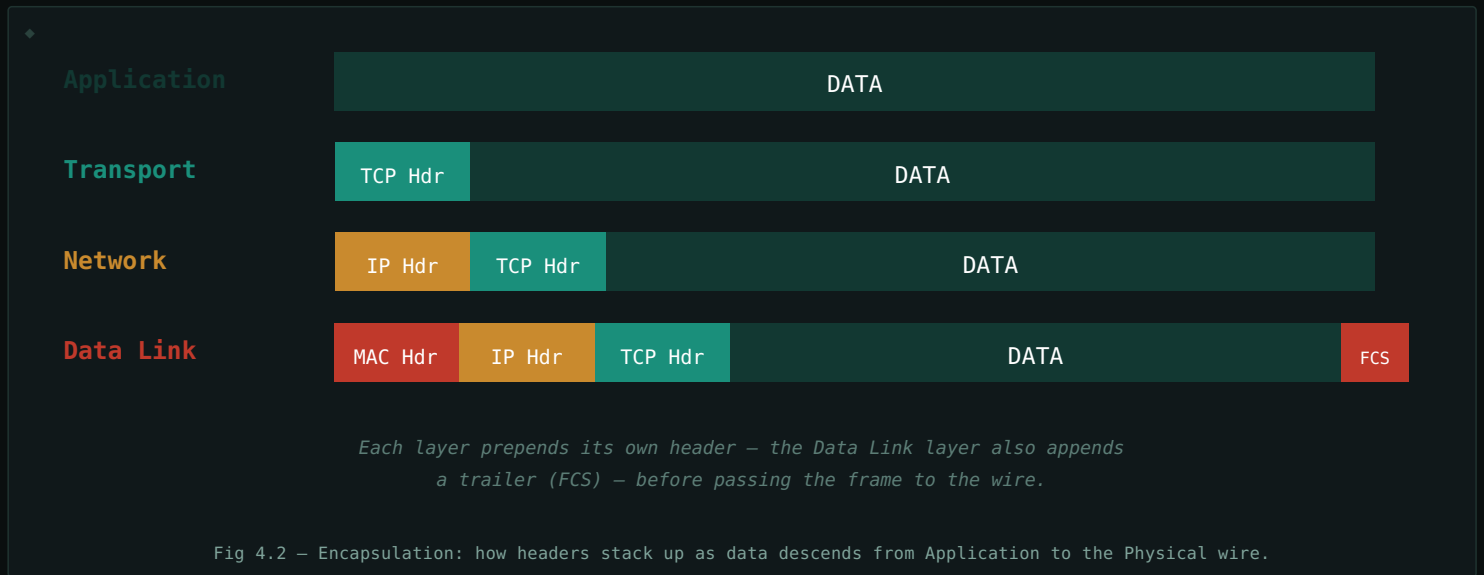
The **Open Systems Interconnection (OSI)** model, published by ISO, is a conceptual framework that splits network communication into **seven layers**. Each layer has a specific job and communicates only with the layers directly above and below it. It is a teaching and design model – real-world stacks like TCP/IP don't implement it exactly, but understanding OSI makes every other networking concept easier to place.



4.1 LAYER-BY-LAYER BREAKDOWN

#	LAYER	PDU	FUNCTION	EXAMPLES
7	Application	Data	Interfaces directly with end-user software; provides network services to applications.	HTTP, FTP, SMTP, DNS
6	Presentation	Data	Translates, encrypts, and compresses data so the application layer can use it – the "translator."	SSL/TLS, JPEG, ASCII, encryption
5	Session	Data	Establishes, manages, and terminates sessions (dialogues) between two hosts.	NetBIOS, RPC, PPTP session setup
4	Transport	Segment	End-to-end delivery, reliability, flow control, and error recovery between hosts.	TCP, UDP
3	Network	Packet	Logical addressing (IP) and routing of packets across different networks.	IP, ICMP, routers
2	Data Link	Frame	Physical/MAC addressing, framing, and error detection on a single link.	Ethernet, Wi-Fi (802.11), switches
1	Physical	Bit	Transmission of raw bits over the physical medium – voltages, light pulses, radio waves.	Cables, connectors, hubs, NIC

Encapsulation: As data moves down the stack on the sender's side, each layer wraps (encapsulates) the data from the layer above with its own header (and sometimes trailer). On the receiver's side, each layer strips off its corresponding header as data moves back up – this process is called **de-encapsulation**.



4.2 DATA LINK LAYER SUBLAYERS

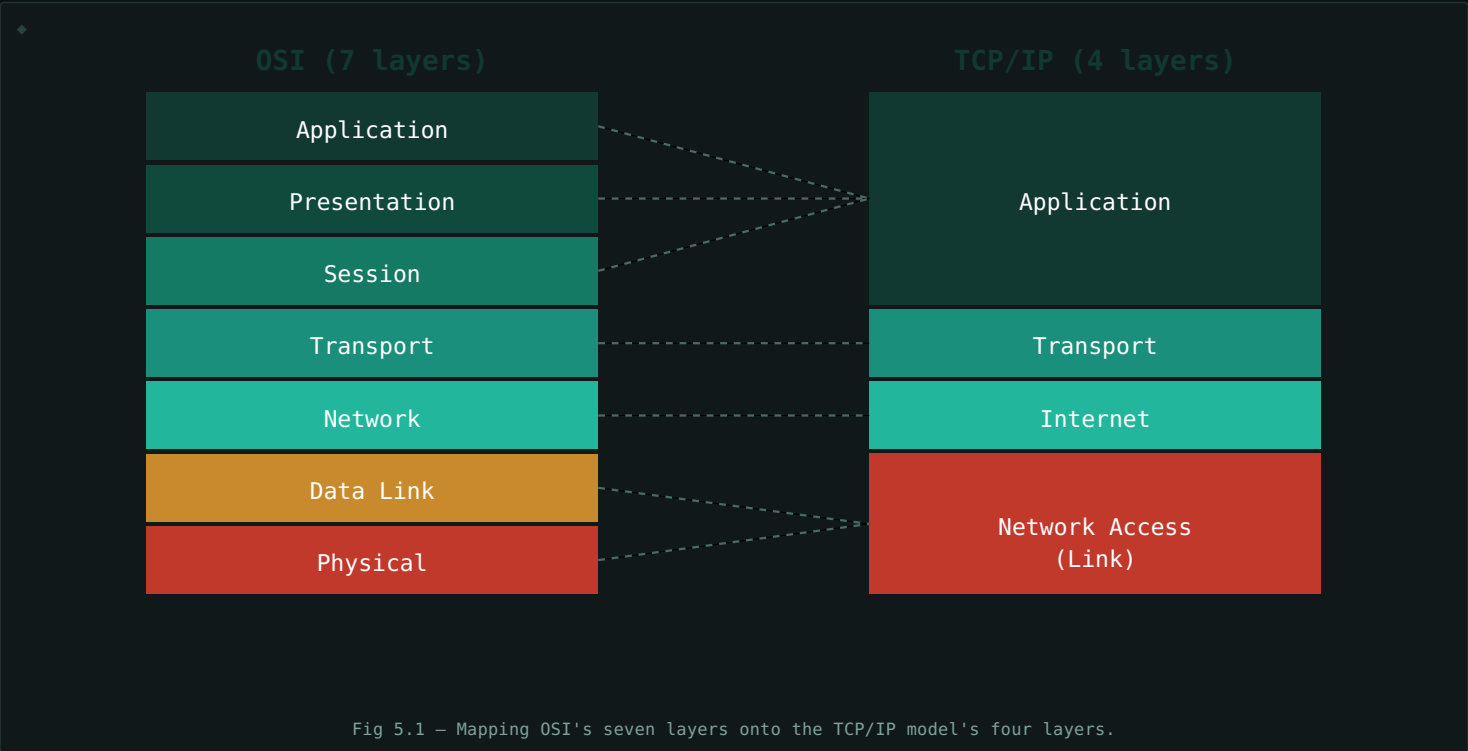
The Data Link layer is often split into two sublayers defined by the IEEE 802 standards: **LLC (Logical Link Control)**, which handles flow/error control and talks to the Network layer above, and **MAC (Media Access Control)**, which handles addressing and controls access to the shared physical medium.

SUBLAYER	FACES	RESPONSIBILITY
LLC	Network layer (above)	Multiplexes protocols, provides flow control and optional reliability on top of MAC.
MAC	Physical layer (below)	Frames data, assigns/reads MAC addresses, and arbitrates access to the shared medium (e.g. CSMA/CD, CSMA/CA).

This split is why a single physical Ethernet or Wi-Fi card can support multiple upper-layer protocols simultaneously – the LLC sublayer's job is exactly that kind of traffic multiplexing, identified in a frame via the EtherType field.

The TCP/IP Model

The **TCP/IP model** (also called the Internet Protocol Suite) is the practical, four-layer model that the real Internet actually runs on. It predates OSI and is less rigid – several OSI layers are collapsed together.



5.1 THE FOUR LAYERS

LAYER	CORRESPONDS TO OSI	KEY PROTOCOLS
Application	Application + Presentation + Session	HTTP/HTTPS, FTP, SMTP, DNS, Telnet, SSH
Transport	Transport	TCP, UDP
Internet	Network	IP (IPv4/IPv6), ICMP, ARP
Network Access (Link)	Data Link + Physical	Ethernet, Wi-Fi, PPP

5.2 OSI VS TCP/IP – KEY DIFFERENCES

ASPECT	OSI MODEL	TCP/IP MODEL
Layers	7	4 (sometimes shown as 5)
Development	Theoretical, ISO standard	Practical, grew from ARPANET
Usage	Reference/teaching model	Actually implemented on the Internet
Approach	Protocol-independent	Protocol-dependent (built around TCP & IP)
Layer coupling	Strict, each layer clearly separated	Looser, some layers combined

Transmission Media

Transmission media are the physical or wireless pathways that carry signals between devices. They fall into two broad categories: **guided (wired)** media, where signals travel along a solid conductor, and **unguided (wireless)** media, where signals radiate through free space.

6.1 GUIDED MEDIA

Twisted Pair Cable

Two insulated copper wires twisted around each other to reduce electromagnetic interference (crosstalk). Comes in **UTP** (Unshielded Twisted Pair) and **STP** (Shielded Twisted Pair). Cheap and widely used for LANs (Cat5e, Cat6, Cat6a cabling).

Coaxial Cable

A central copper conductor surrounded by insulation, a metallic shield, and an outer jacket. More resistant to interference than twisted pair; historically used for cable TV and older Ethernet (10BASE2/5), still used for cable internet (DOCSIS).

Fiber Optic Cable

Transmits data as pulses of light through a glass or plastic core. Immune to electromagnetic interference, supports very high bandwidth over long distances. **Single-mode fiber** uses a narrow core for long-haul, high-speed links; **multi-mode fiber** uses a wider core for shorter, cheaper runs.



6.2 UNGUIDED (WIRELESS) MEDIA

- **Radio Waves** – omnidirectional, penetrate walls; used for Wi-Fi, FM radio, cordless phones.
- **Microwaves** – line-of-sight, directional; used for point-to-point links and satellite communication.
- **Infrared** – short range, cannot pass through walls; used for remote controls and short-range device links.

6.3 MEDIA COMPARISON

MEDIUM	BANDWIDTH	INTERFERENCE IMMUNITY	COST	TYPICAL RANGE
Twisted Pair (Cat6)	Up to 10 Gbps	Low-Medium	Low	~100 m
Coaxial	Up to ~1 Gbps	Medium	Medium	~500 m
Fiber Optic	Multiple Tbps (theoretical)	Very High	High	Kilometers
Radio / Wi-Fi	Up to ~9.6 Gbps (Wi-Fi 6E)	Low	Low	Tens of meters

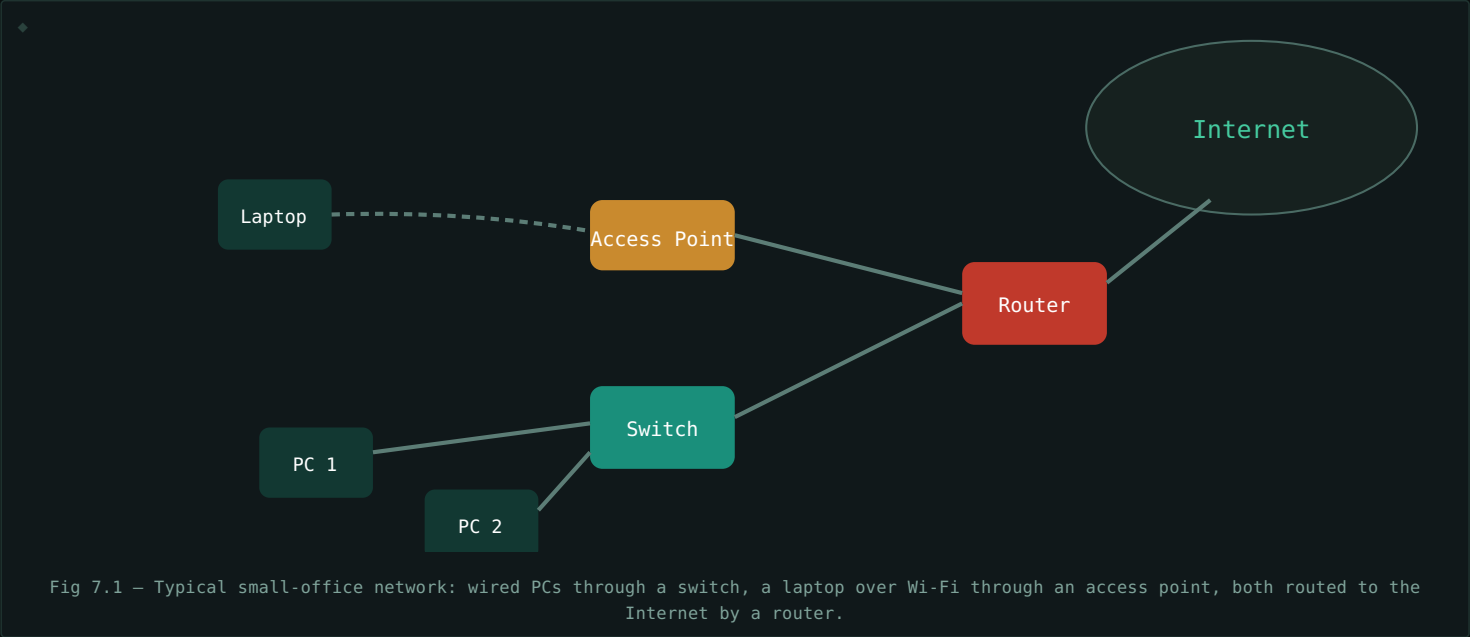
Networking Devices

Intermediary devices connect, extend, filter, and direct traffic within and between networks. Each operates at a particular OSI layer.

7.1 DEVICE REFERENCE TABLE

DEVICE	OSI LAYER	FUNCTION
Repeater	Physical (L1)	Regenerates a weak signal to extend cable distance.
Hub	Physical (L1)	Multi-port repeater; broadcasts incoming data to all ports (no intelligence, causes collisions).
Bridge	Data Link (L2)	Connects two LAN segments and filters traffic using MAC addresses.
Switch	Data Link (L2)	Multi-port bridge; forwards frames only to the port of the intended MAC address, reducing collisions.
Router	Network (L3)	Forwards packets between different networks based on IP addresses; connects LANs to WANs/the Internet.
Gateway	All layers	Translates between completely different protocol stacks (e.g., connecting a LAN to a mainframe network).
Modem	Physical (L1)	MOdulates/DEModulates digital signals to/from analog signals for transmission over telephone or cable lines.
Access Point (AP)	Data Link (L2)	Allows wireless devices to connect to a wired network using Wi-Fi.
Firewall	Varies (L3–L7)	Filters traffic based on security rules; can be hardware or software.

Networking Devices



7.2 HUB VS SWITCH VS ROUTER

FEATURE	HUB	SWITCH	ROUTER
OSI Layer	1 (Physical)	2 (Data Link)	3 (Network)
Addressing used	None	MAC address	IP address
Traffic handling	Broadcasts to all ports	Forwards to specific port	Routes between networks
Collision domain	Single, shared	Separate per port	Separate per interface
Intelligence	None	MAC address table	Routing table, NAT, DHCP possible

Switching Techniques

Switching determines how data moves through intermediate nodes from source to destination across a network made of many interconnected links.

8.1 CIRCUIT SWITCHING

A dedicated physical path (circuit) is established between sender and receiver for the entire duration of the communication, then torn down when finished. Guarantees consistent bandwidth but wastes capacity when the line is idle. Classic example: the traditional telephone network (PSTN).

8.2 PACKET SWITCHING

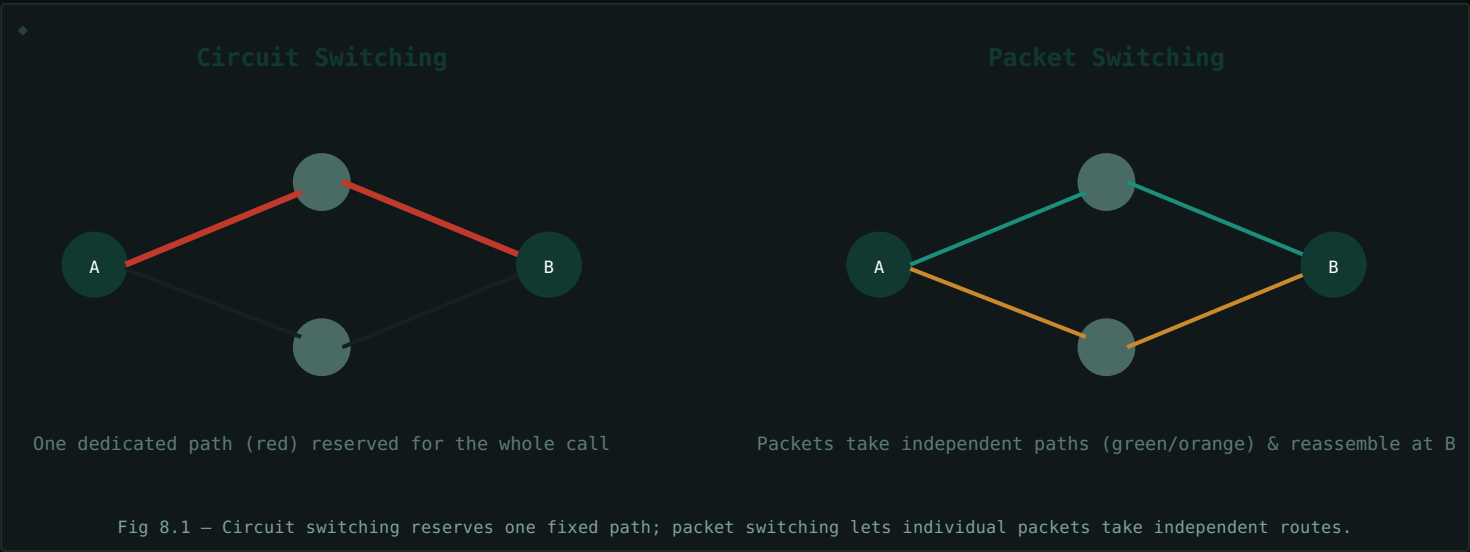
Data is broken into small units called **packets**, each carrying source/destination addressing. Packets are routed independently and may take different paths, then reassembled at the destination. Much more efficient use of bandwidth, since the same links are shared by many conversations. This is how the Internet works.

Datagram (Connectionless) vs Virtual Circuit (Connection-Oriented) Packet Switching

- **Datagram approach** – each packet is routed independently based on current network conditions (e.g., IP). No setup phase; packets may arrive out of order.
- **Virtual circuit approach** – a logical path is set up before data transfer, and all packets follow that same path (e.g., ATM, MPLS, or TCP's logical connection over IP).

8.3 MESSAGE SWITCHING

An older store-and-forward technique where an entire message is sent to an intermediate node, stored completely, and then forwarded to the next node once the link is free. No dedicated path is required, but latency is high and it demands large storage at each node – mostly of historical interest today (early telegraph/email-relay systems).



8.4 COMPARISON

ASPECT	CIRCUIT SWITCHING	PACKET SWITCHING
Path	Fixed, dedicated	Dynamic, shared
Setup delay	Yes, before transmission	No (datagram) / minimal (virtual circuit)
Bandwidth use	Can be wasteful when idle	Efficient, statistically multiplexed
Reliability	Guaranteed once connected	Best-effort; may need retransmission
Example	PSTN telephone calls	The Internet (IP)

IP Addressing & Subnetting

9.1 WHAT IS AN IP ADDRESS?

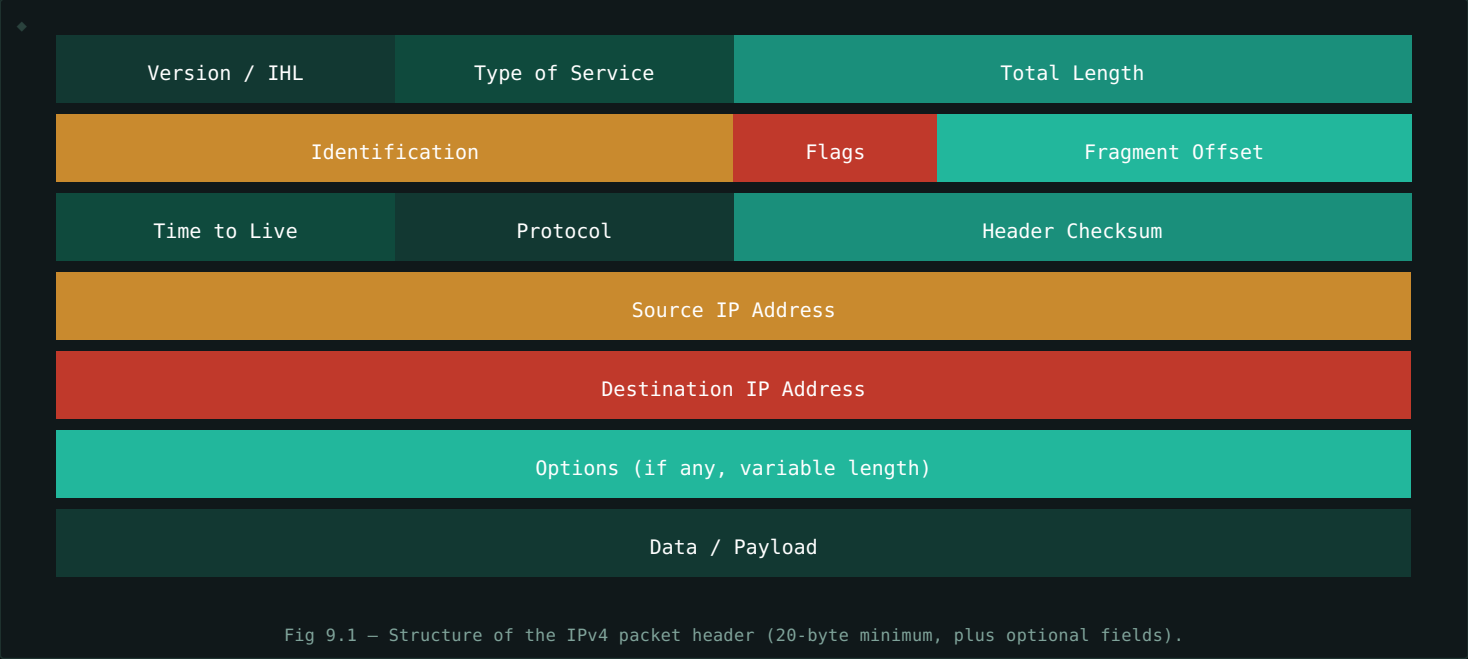
An **IP address** is a unique logical identifier assigned to every device on a network, allowing packets to find their way to the correct host across interconnected networks. **IPv4** addresses are 32-bit numbers written as four decimal octets (e.g. `192.168.1.10`); **IPv6** addresses are 128-bit, written in hexadecimal groups (e.g. `2001:0db8:85a3::8a2e:0370:7334`) to solve IPv4's address exhaustion problem.

9.2 IPV4 ADDRESS CLASSES

CLASS	LEADING BITS	RANGE (FIRST OCTET)	DEFAULT MASK	TYPICAL USE
A	0	1 – 126	255.0.0.0 (/8)	Very large networks
B	10	128 – 191	255.255.0.0 (/16)	Medium-large networks
C	110	192 – 223	255.255.255.0 (/24)	Small networks
D	1110	224 – 239	–	Multicast
E	1111	240 – 255	–	Reserved / experimental

Note: `127.0.0.0/8` is reserved for loopback (e.g. `127.0.0.1` = localhost). Private (non-routable) ranges include `10.0.0.0/8`, `172.16.0.0/12`, and `192.168.0.0/16`.

9.3 THE IPV4 HEADER



9.4 SUBNETTING

Subnetting divides one large network into smaller sub-networks (subnets) by borrowing bits from the host portion of an address for the network portion. This improves address utilization, reduces broadcast traffic, and lets administrators segment networks logically (e.g., separating departments).

CIDR notation – modern networks use **Classless Inter-Domain Routing**, written as `address/prefix-length` (e.g. `192.168.1.0/26`), instead of rigid class-based masks. The `/n` tells you how many leading bits belong to the network portion.

Worked Example

Suppose we're given the network `192.168.1.0/24` and need to split it into 4 equal subnets.

- 1. We need to borrow 2 host bits to create 4 subnets ($2^2 = 4$), giving a new mask of `/26` (255.255.255.192).
- 2. Each subnet then has $2^6 = 64$ addresses (62 usable after network + broadcast).

SUBNET	NETWORK ADDRESS	USABLE HOST RANGE	BROADCAST ADDRESS
1	192.168.1.0	192.168.1.1 – .62	192.168.1.63
2	192.168.1.64	192.168.1.65 – .126	192.168.1.127
3	192.168.1.128	192.168.1.129 – .190	192.168.1.191
4	192.168.1.192	192.168.1.193 – .254	192.168.1.255

9.5 IPV4 VS IPV6

ASPECT	IPV4	IPV6
Address length	32 bits	128 bits
Address space	~4.3 billion addresses	~340 undecillion addresses
Notation	Dotted decimal (192.168.1.1)	Hexadecimal, colon-separated
Header	Variable length, includes checksum	Fixed 40-byte header, no checksum (faster routing)
Configuration	Manual or DHCP	Supports stateless auto-configuration (SLAAC)
Built-in security	Optional (IPsec add-on)	IPsec support built into the spec

Core Protocols

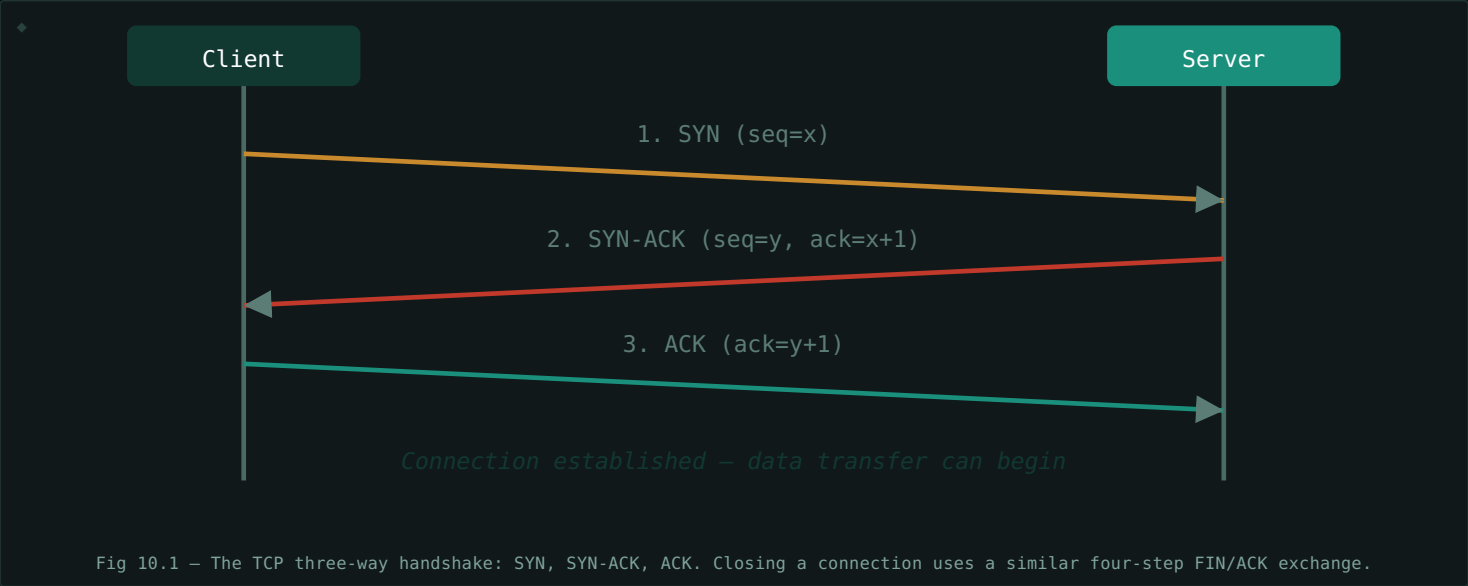
10.1 TCP VS UDP

Both are Transport-layer protocols that deliver data between applications using **port numbers**, but they take opposite approaches to reliability.

FEATURE	TCP (TRANSMISSION CONTROL PROTOCOL)	UDP (USER DATAGRAM PROTOCOL)
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Guaranteed delivery, retransmits lost data	Best-effort, no retransmission
Ordering	Preserves byte order	No ordering guarantee
Speed	Slower (overhead of acknowledgments)	Faster, minimal overhead
Header size	20 bytes	8 bytes
Use cases	Web browsing, email, file transfer	Video streaming, VoIP, DNS, online gaming

10.2 THE TCP THREE-WAY HANDSHAKE

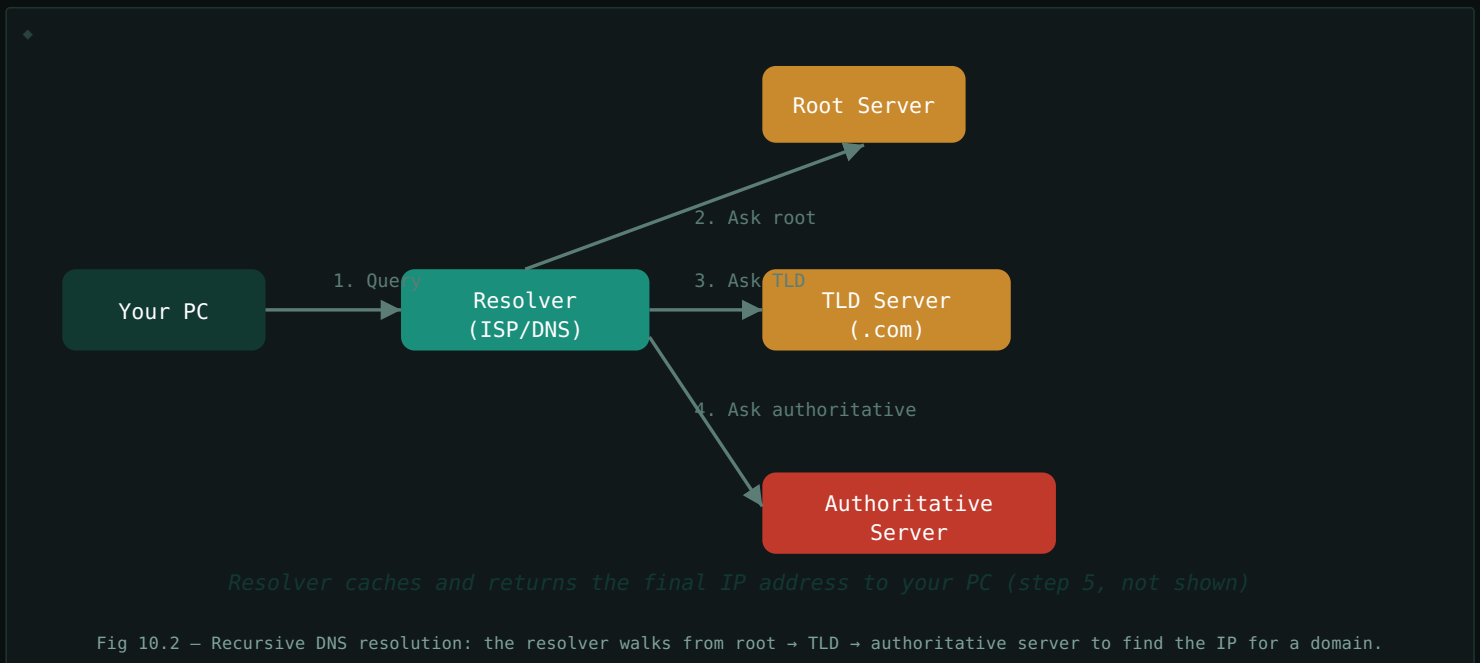
Before exchanging data, TCP establishes a connection using three messages:



10.3 WELL-KNOWN APPLICATION LAYER PROTOCOLS

PROTOCOL	PORT	TRANSPORT	PURPOSE
HTTP	80	TCP	Transfers web pages (unencrypted)
HTTPS	443	TCP	HTTP secured with TLS/SSL encryption
FTP	20/21	TCP	File transfer between client and server
SMTP	25 (587 for submission)	TCP	Sends outgoing email
POP3	110	TCP	Downloads email to a client, usually removing it from the server
IMAP	143	TCP	Syncs email across devices, keeping mail on the server
DNS	53	UDP (TCP for large queries)	Translates domain names to IP addresses
DHCP	67/68	UDP	Automatically assigns IP addresses to devices
SSH	22	TCP	Secure encrypted remote login/command execution
Telnet	23	TCP	Unencrypted remote login (largely obsolete for security reasons)
SNMP	161/162	UDP	Monitors and manages network devices

10.4 HOW DNS RESOLUTION WORKS



10.5 DHCP – AUTOMATIC IP ASSIGNMENT

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, gateways, and DNS servers to devices joining a network, using a four-step process remembered by the mnemonic **DORA**:

1. **Discover** – client broadcasts a request for an IP address.
2. **Offer** – DHCP server responds with an available IP address.
3. **Request** – client formally requests the offered address.
4. **Acknowledge** – server confirms and leases the address to the client.

Error Detection & Correction

Electrical noise, interference, and attenuation can flip bits during transmission. Networks use mathematical techniques to detect – and sometimes correct – these errors.

11.1 PARITY CHECK

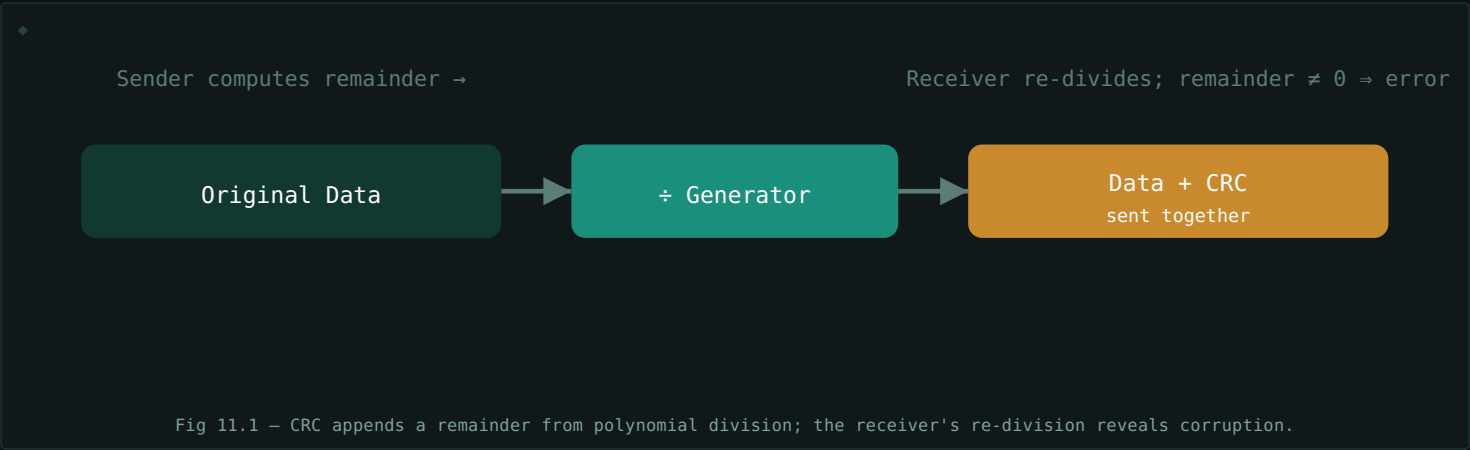
The simplest method: an extra **parity bit** is added so the total number of 1s in the data is even (even parity) or odd (odd parity). The receiver recomputes parity and flags a mismatch. It can only reliably catch an odd number of bit errors – it misses 2-bit errors.

11.2 CHECKSUM

The sender breaks data into fixed-size sections, adds them together, and sends the complement of the sum (checksum) alongside the data. The receiver performs the same sum, including the checksum; if the result isn't all 1s (or 0, depending on convention), an error is flagged. Used in IP, TCP, and UDP headers.

11.3 CYCLIC REDUNDANCY CHECK (CRC)

A far more powerful technique used at the Data Link layer (e.g., Ethernet frames). The sender treats the data as a large binary number and divides it by a fixed generator polynomial; the remainder becomes the CRC value appended to the frame. The receiver performs the same division – a non-zero remainder means the frame is corrupted. CRC reliably detects burst errors and is far stronger than simple checksums.



11.4 HAMMING CODE (ERROR CORRECTION)

Unlike parity or CRC (which only detect errors), **Hamming code** can locate and correct single-bit errors without retransmission. It works by inserting redundant parity bits at positions that are powers of two (1, 2, 4, 8...), each covering a specific overlapping set of data bits. If a bit flips, the pattern of failed parity checks points directly to the faulty bit position, which is then flipped back. It's used in ECC RAM and satellite communication, where retransmission is costly or impossible.

11.5 COMPARISON

TECHNIQUE	DETECTS	CORRECTS	OVERHEAD	COMMON USE
Parity Bit	Single-bit errors (odd count)	No	Very low	Simple serial links
Checksum	Most errors, weak against some patterns	No	Low	IP, TCP, UDP headers
CRC	Burst errors, very reliable	No	Moderate	Ethernet frames, ZIP files
Hamming Code	Single & some double-bit errors	Single-bit errors	Higher	ECC memory, satellite links

Flow Control & Congestion Control

12.1 FLOW CONTROL – PROTECTING THE RECEIVER

Flow control prevents a fast sender from overwhelming a slow receiver's buffer.

Stop-and-Wait

The sender transmits one frame, then waits for an acknowledgment (ACK) before sending the next. Simple but inefficient – the link sits idle while waiting for the ACK.

Sliding Window

The sender can transmit multiple frames (up to a negotiated **window size**) before needing an acknowledgment, greatly improving throughput. TCP uses a dynamically-adjusted sliding window advertised by the receiver.



12.2 CONGESTION CONTROL – PROTECTING THE NETWORK

Congestion control prevents too much traffic from overwhelming the network itself (routers, links), not just the receiver. TCP implements several algorithms:

- **Slow Start** – the sender begins with a small congestion window and doubles it every round-trip until reaching a threshold or detecting loss.
- **Congestion Avoidance** – once near the threshold, the window grows linearly instead of exponentially, to probe capacity more cautiously.
- **Fast Retransmit** – if the sender receives three duplicate ACKs, it assumes a packet was lost and retransmits immediately, without waiting for a timeout.
- **Fast Recovery** – after a fast retransmit, the window is cut (not reset to the minimum), letting the connection resume near current throughput instead of starting over.

Flow control vs congestion control: flow control is a private negotiation between one sender and one receiver about buffer space; congestion control is the network-wide concern of avoiding overload across many senders sharing the same links.

Routing Algorithms & Protocols

Routing is the process routers use to decide the best path for a packet to reach its destination network. Routers build and maintain a **routing table** using either manual configuration or dynamic routing protocols.

13.1 STATIC VS DYNAMIC ROUTING

ASPECT	STATIC ROUTING	DYNAMIC ROUTING
Configuration	Manually entered by an administrator	Automatically learned via routing protocols
Adapts to change	No – must be updated manually	Yes – recalculates when topology changes
Overhead	None (no protocol traffic)	Some CPU/bandwidth used for updates
Best for	Small, stable networks	Large, changing networks

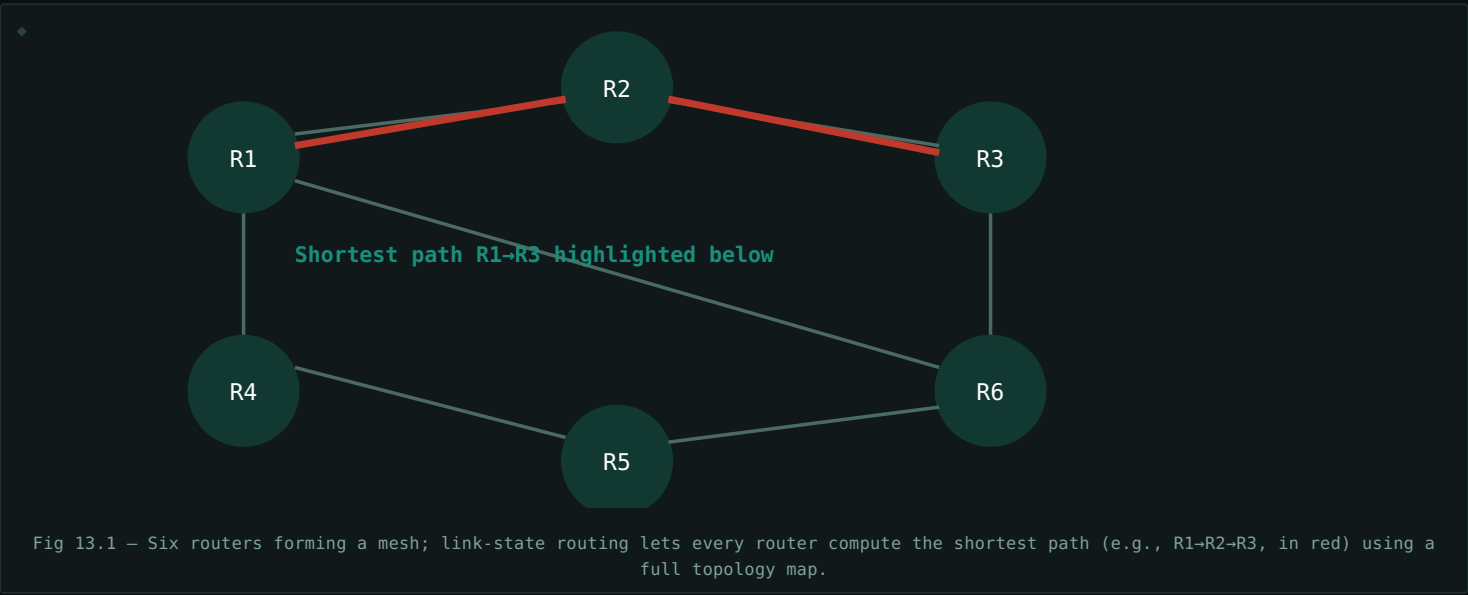
13.2 DISTANCE-VECTOR VS LINK-STATE ROUTING

Distance-Vector (e.g. RIP)

Each router shares its entire routing table with directly-connected neighbors periodically. Routers know the *distance* (hop count) to a destination and the *direction* (which neighbor) to send through, but not the network's overall topology – described by the phrase "routing by rumor."

Link-State (e.g. OSPF)

Each router builds a complete map of the network by flooding information about its direct links to all other routers, then independently calculates the shortest path to every destination using an algorithm like **Dijkstra's**. More accurate and faster to converge than distance-vector, at the cost of more processing and memory.



13.3 COMMON ROUTING PROTOCOLS

PROTOCOL	TYPE	METRIC	SCOPE
RIP	Distance-Vector	Hop count (max 15)	Small networks
OSPF	Link-State	Cost (based on bandwidth)	Enterprise networks (interior)
EIGRP	Hybrid (Cisco proprietary)	Bandwidth, delay, reliability, load	Enterprise networks
BGP	Path-Vector	AS-path attributes/policy	Between ISPs – the Internet's backbone protocol

Wireless Networking

14.1 WI-FI (IEEE 802.11)

Wi-Fi lets devices connect to a LAN without cables, operating primarily in the 2.4 GHz and 5 GHz (and newer 6 GHz) bands.

STANDARD	MARKETING NAME	MAX THEORETICAL SPEED	BAND
802.11n	Wi-Fi 4	600 Mbps	2.4 / 5 GHz
802.11ac	Wi-Fi 5	3.5 Gbps	5 GHz
802.11ax	Wi-Fi 6 / 6E	9.6 Gbps	2.4 / 5 / 6 GHz
802.11be	Wi-Fi 7	46 Gbps (theoretical)	2.4 / 5 / 6 GHz

14.2 WIRELESS SECURITY MODES

Older **WEP** encryption is broken and should never be used. **WPA2** (using AES-CCMP) has been the practical standard for years; **WPA3** adds stronger encryption and protects against offline password-guessing attacks (via Simultaneous Authentication of Equals). As a SOC-facing note: an open or WEP-secured access point should always be flagged as a high-priority finding in a wireless assessment.

14.3 BLUETOOTH

A short-range (typically under 10 m) PAN technology operating in the 2.4 GHz band, designed for low-power device pairing – headphones, keyboards, fitness trackers. Bluetooth Low Energy (BLE) trades speed for dramatically lower power consumption, making it the standard for IoT and wearables.

14.4 CELLULAR NETWORKS

GENERATION	KEY FEATURE	TYPICAL SPEED
3G	Mobile data, basic browsing	~2 Mbps
4G / LTE	Mobile broadband, VoLTE	~100 Mbps
5G	Ultra-low latency, network slicing, massive IoT	Up to ~10 Gbps (peak)

14.5 WIRELESS VS WIRED – TRADEOFFS

ASPECT	WIRED	WIRELESS
Speed & Latency	Generally higher, more stable	Lower, variable (interference, distance)
Mobility	None – fixed location	High – devices can roam
Security	Physical access required to tap	Signal can be intercepted over the air
Installation	Labor-intensive cabling	Quick to deploy

Network Security Fundamentals

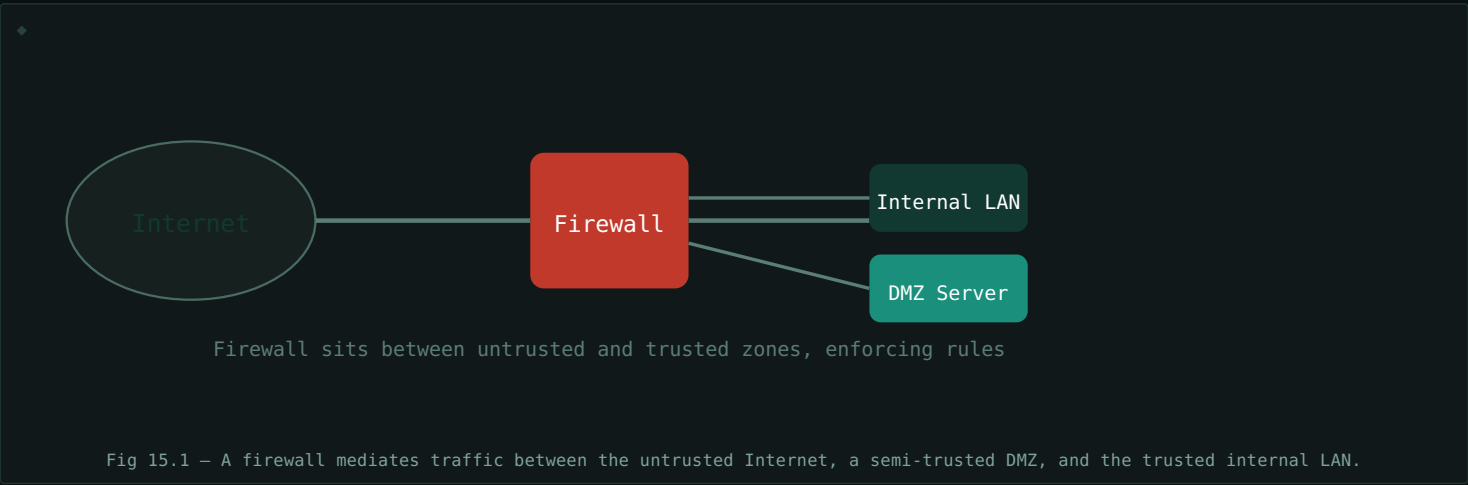
15.1 THE CIA TRIAD

Nearly all network security goals map back to three core properties:

- **Confidentiality** – only authorized parties can read the data (achieved via encryption, access control).
- **Integrity** – data isn't altered in transit without detection (achieved via hashing, digital signatures, CRC).
- **Availability** – systems and data remain accessible when needed (protected against DoS/DDoS attacks, achieved via redundancy).

15.2 FIREWALLS

A firewall inspects and filters traffic based on rules – source/destination IP, port, protocol, or, for next-generation firewalls, application signatures and content. Firewalls can be **packet-filtering** (rule-based on headers only), **stateful** (tracks connection state), or **application-layer / proxy** (inspects payload content).



15.3 VPNS (VIRTUAL PRIVATE NETWORKS)

A VPN creates an encrypted **tunnel** across an untrusted network (like the public Internet), making remote traffic appear as if it originates from inside a private network. Common protocols include **IPsec** (network-layer encryption, often site-to-site) and **SSL/TLS VPNs** (application-layer, often used for remote-access clients like OpenVPN or WireGuard).

15.4 COMMON ATTACK TYPES (BLUE-TEAM REFERENCE)

ATTACK	LAYER / AREA	DESCRIPTION
ARP Spoofing	Data Link	Attacker sends forged ARP replies to associate their MAC with another host's IP, enabling man-in-the-middle interception.
DNS Spoofing / Cache Poisoning	Application	Injects false DNS responses to redirect victims to malicious sites.
SYN Flood	Transport	Attacker sends many SYN packets without completing the handshake, exhausting the server's connection queue (a form of DoS).
Man-in-the-Middle (MITM)	Multiple	Attacker secretly intercepts and possibly alters communication between two parties.
DDoS	Multiple / Network	Coordinated flood of traffic from many sources to exhaust target resources or bandwidth.
Packet Sniffing	Data Link / Network	Passive capture of unencrypted traffic on a shared or compromised segment.

Defense in depth: no single control (firewall, encryption, IDS/IPS, patching, monitoring) is sufficient alone – layered, overlapping defenses are what make networks resilient to real-world attackers.

15.5 IDS VS IPS

ASPECT	IDS (INTRUSION DETECTION SYSTEM)	IPS (INTRUSION PREVENTION SYSTEM)
Placement	Out-of-band (monitors a copy of traffic)	Inline (traffic passes through it)
Action	Detects and alerts	Detects and actively blocks/drops
Risk	No impact on traffic flow if it fails	A false positive can block legitimate traffic

Glossary of Key Terms

TERM	DEFINITION
Bandwidth	The maximum rate of data transfer across a link, usually measured in bits per second.
Latency	The time delay between sending and receiving data, often measured as round-trip time (RTT).
Throughput	The actual rate of successful data delivery, which is often lower than the theoretical bandwidth.
Jitter	Variation in packet delay, especially disruptive for real-time voice/video traffic.
Packet	A formatted unit of data carrying both a header (addressing/control info) and a payload.
MAC Address	A globally unique 48-bit hardware address burned into a network interface card.
Port Number	A 16-bit value identifying a specific process or service on a host (e.g., port 443 for HTTPS).
NAT	Network Address Translation – maps private internal IPs to a public IP for Internet access.
Broadcast	A transmission sent to every device on a network segment.
Multicast	A transmission sent to a specific group of interested receivers, not everyone.
Unicast	A transmission sent from one sender to exactly one receiver.
Bandwidth-Delay Product	$\text{Bandwidth} \times \text{RTT}$ – the amount of data "in flight" on a link at any moment.
MTU	Maximum Transmission Unit – the largest packet size a link can carry without fragmentation.
Loopback Address	127.0.0.1 (IPv4) or ::1 (IPv6) – used by a host to test its own network stack.
Default Gateway	The router a device sends traffic to when the destination is outside its local subnet.
ARP	Address Resolution Protocol – maps a known IP address to its corresponding MAC address on a LAN.
Ping / ICMP	A diagnostic tool/protocol used to test reachability and measure round-trip latency between hosts.
Full Duplex	Communication where both ends can send and receive simultaneously.
Half Duplex	Communication where only one end can transmit at a time.

End of guide. This book covers the foundational theory behind computer networking – from cabling and topologies up through routing, protocols, and security. For deeper mastery, pair this with hands-on practice: build a small home lab, capture traffic with Wireshark, and configure a router/switch in a simulator like Cisco Packet Tracer or GNS3.